

Problem-Solution Fit

Date	28 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Students studying difficult subjects online - School and college students preparing for exams - Self-learners using e-learning platforms - Students who struggle to stay motivated	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited access to personal tutors - Basic laptops/phones with average internet - Time constraints between classes and exams - Limited awareness of their own learning emotions	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions: generic e-learning platforms, YouTube tutorials, static FAQ chatbots Pros: - Widely accessible - Free or low-cost content Cons: - Not personalized to how the student feels - No awareness of confusion or frustration - One-size-fits-all explanations
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Difficulty understanding certain topics (Very Frequent) - Loss of motivation while studying (Frequent) - Unable to identify why they're stuck (Frequent) - Anxiety before exams (Occasional)	9. PROBLEM ROOT / CAUSE [RC] - Lack of emotional awareness in existing platforms - No mechanism to detect confusion or frustration - Generic content regardless of learner's state - Limited feedback on why a topic feels hard	7. BEHAVIOR + ITS INTENSITY [BE] - Re-reads or rewatches the same material (High) - Searches online for simpler explanations (Medium) - Avoids difficult topics until the last moment (Medium)
3. TRIGGERS TO ACT [TR] - Studying for exams or assignments - Repeated failure to understand a concept - Preparing for an upcoming test	10. YOUR SOLUTION [SL] Emotion Detection & Learning Support Engine uses BERT, BiLSTM, and Google Gemini AI to detect students' learning emotions and provide personalized support. - Detects primary and mixed emotions from learning problems - Compares BERT and BiLSTM predictions with confidence scores - Generates personalized study guidance and study plans - Tracks emotion trends through an analytics dashboard	8. CHANNELS OF BEHAVIOR [CH] ONLINE - YouTube - Online forums - E-learning platforms - Social media study groups OFFLINE - Classrooms - Study groups - Tuition centers - Libraries
4. EMOTIONS BEFORE / AFTER [EM] Before: Confused, frustrated, anxious, discouraged After: Confident, motivated, relieved, encouraged		